AI-ML Assignment: Exploratory Data Analysis (EDA) Project

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Exploratory Data Analysis on Hotel Booking Demand Dataset

1. Introduction

Dataset Source

The dataset used in this project is the Hotel Booking Demand Dataset obtained from Kaggle. The dataset contains booking information for City Hotels and Resort Hotels, including customer details, booking characteristics, stay duration, market segments, and pricing information.

Data Dictionary

Column	Description
hotel	Type of hotel (City Hotel or Resort Hotel)
is_canceled	Indicates whether the booking was cancelled
lead_time	Number of days between booking and arrival
arrival_date_year	Year of arrival
arrival_date_month	Month of arrival
stays_in_weekend_nights	Weekend nights booked
stays_in_week_nights	Weekday nights booked
adults	Number of adults
children	Number of children
babies	Number of babies
meal	Type of meal plan selected
country	Customer country
market_segment	Booking market segment
customer_type	Customer type
adr	Average Daily Rate

Business Questions

1. Which hotel type receives more bookings?
2. Does lead time affect booking cancellations?
3. Which market segment contributes the most bookings?
4. Which hotel type generates a higher Average Daily Rate (ADR)?
5. How do guest counts influence booking patterns?

2. Data Cleaning and Integrity Checks

Data Type Verification

The dataset was examined using the .info() function to verify that all columns were loaded with appropriate data types. No major data type conversion was required.

Missing Value Handling

Missing values were identified using the isnull().sum() function. Numerical columns were imputed using mean values, while categorical columns were imputed using mode values.

This approach was selected because removing rows would result in loss of useful information. Mean and mode imputation preserve the dataset while maintaining its statistical properties.

Duplicate Row Removal

Duplicate records were identified using the duplicated() function and removed to ensure data quality and avoid biased analysis.

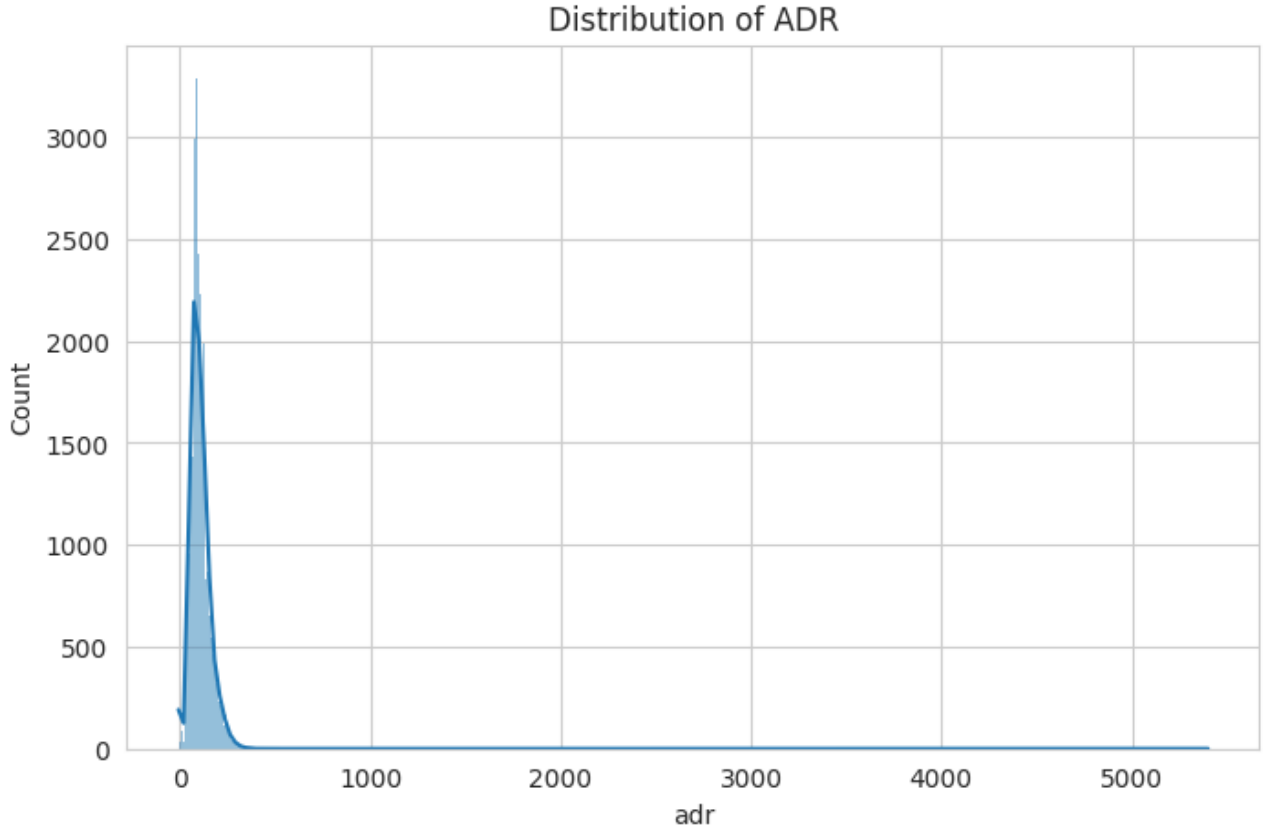
3. Outlier Analysis

Numerical variables were analyzed using skewness metrics, histograms, and boxplots.

The Interquartile Range (IQR) method was applied to detect outliers across continuous variables. Outliers were particularly observed in the Average Daily Rate (ADR) feature.

Instead of removing observations, Winsorization (capping) was applied to reduce the influence of extreme values while preserving valuable information.

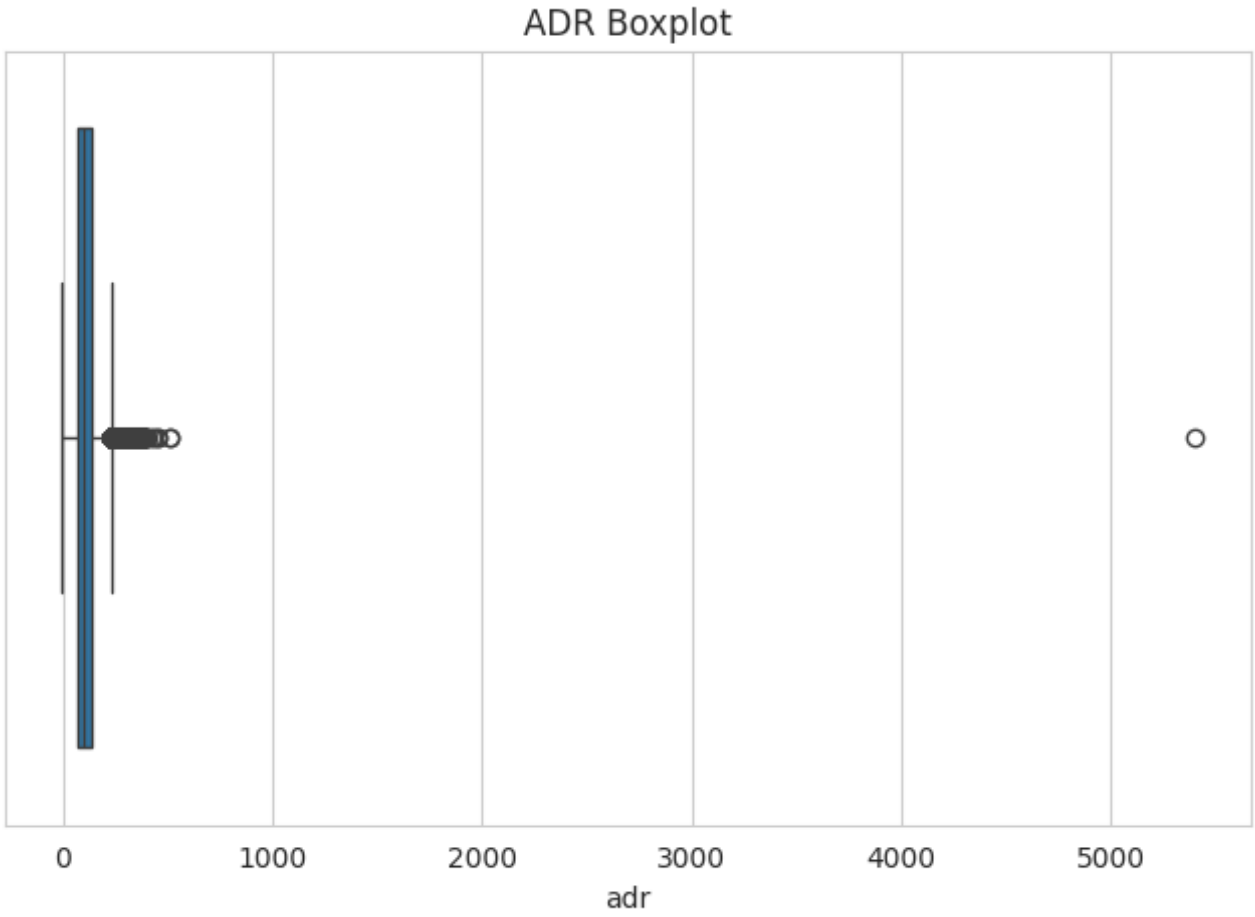
ADR Histogram



Finding

The ADR variable exhibits a right-skewed distribution with a small number of extremly high values.

ADR Boxplot

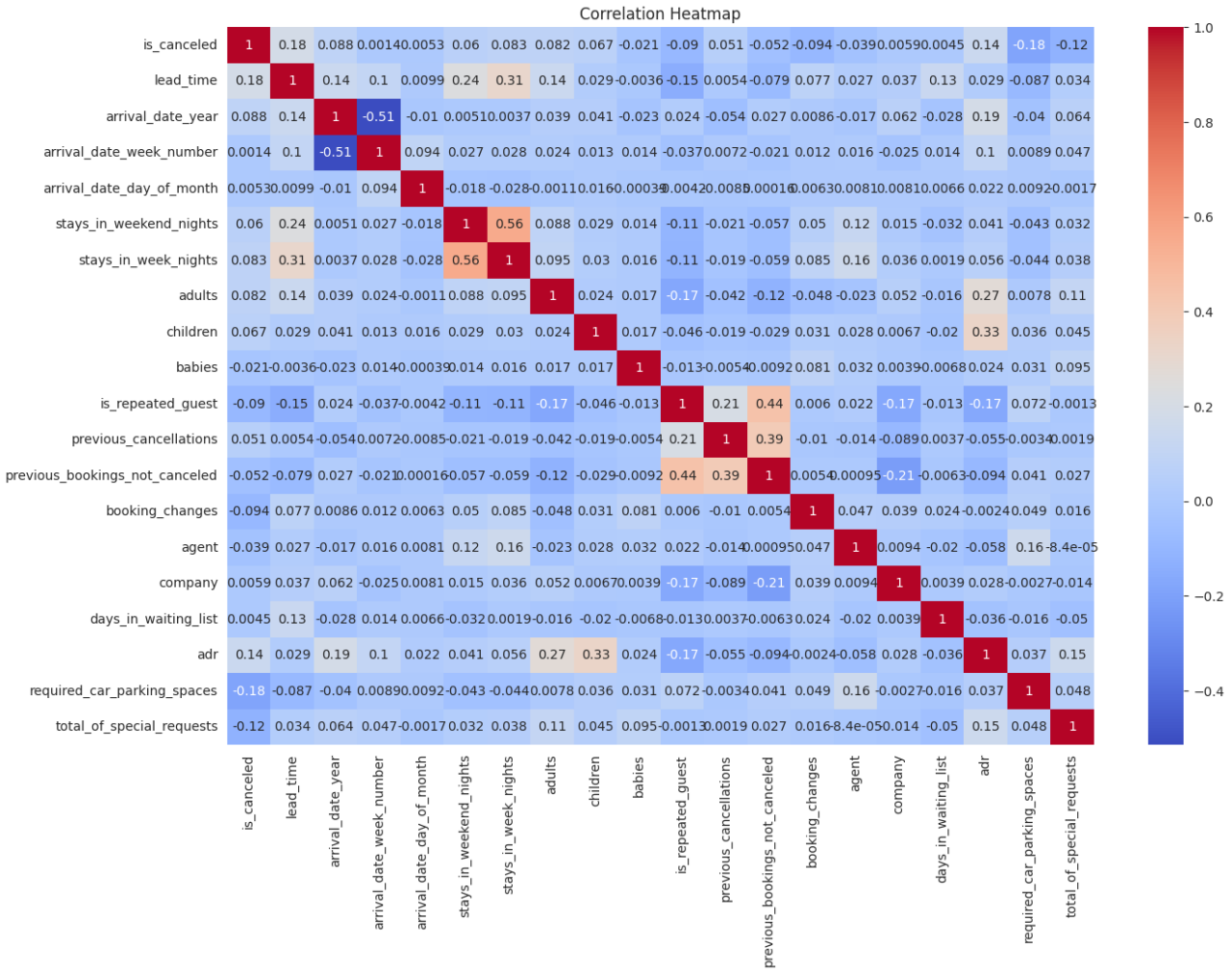


Finding

The boxplot clearly identifies extreme observations outside the upper whisker, indicating the presence of outliers.

4. Bivariate and Multivariate Exploration

Correlation Heatmap



Correlation Analysis

The heatmap was used to analyze relationships among numerical variables. Several variables exhibited moderate positive and negative relationships.

Correlation vs Causation

The observed correlations do not necessarily imply direct causation. Hidden factors such as seasonality, customer preferences, hotel type, and booking behavior may influence the relationships observed in the dataset.

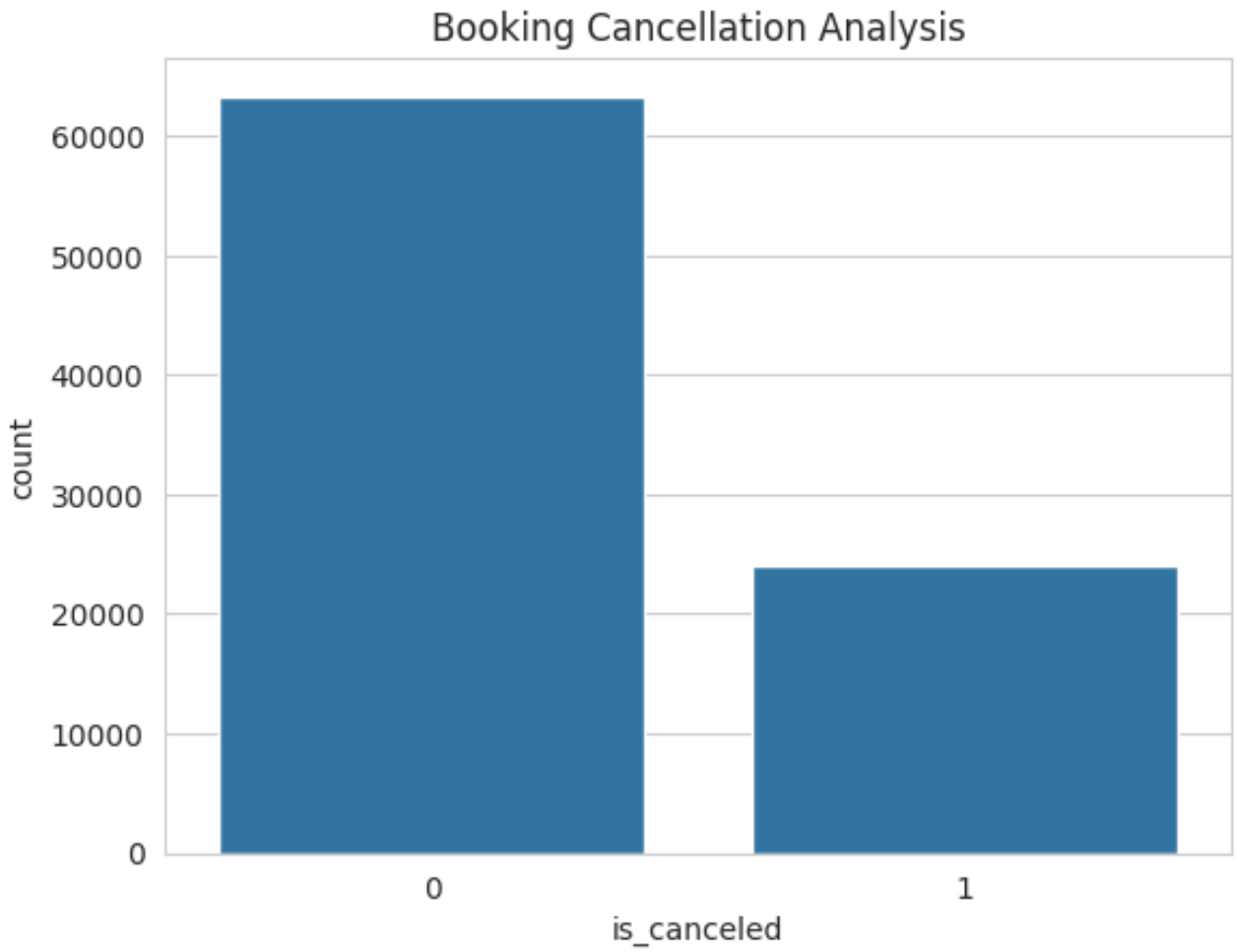
Hotel Type Distribution



Finding

City Hotels received a higher number of bookings compared to Resort Hotels.

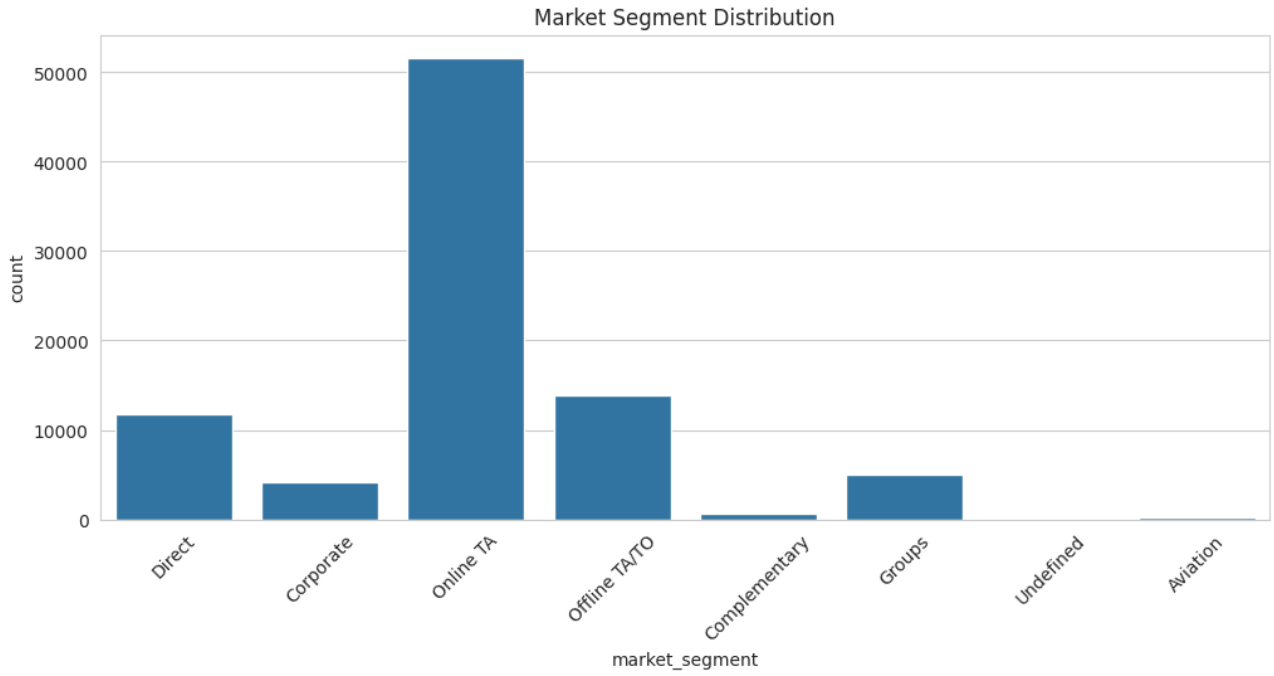
Booking Cancellation Analysis



Finding

A considerable number of bookings were cancelled, indicating that cancellation behavior is an important business factor.

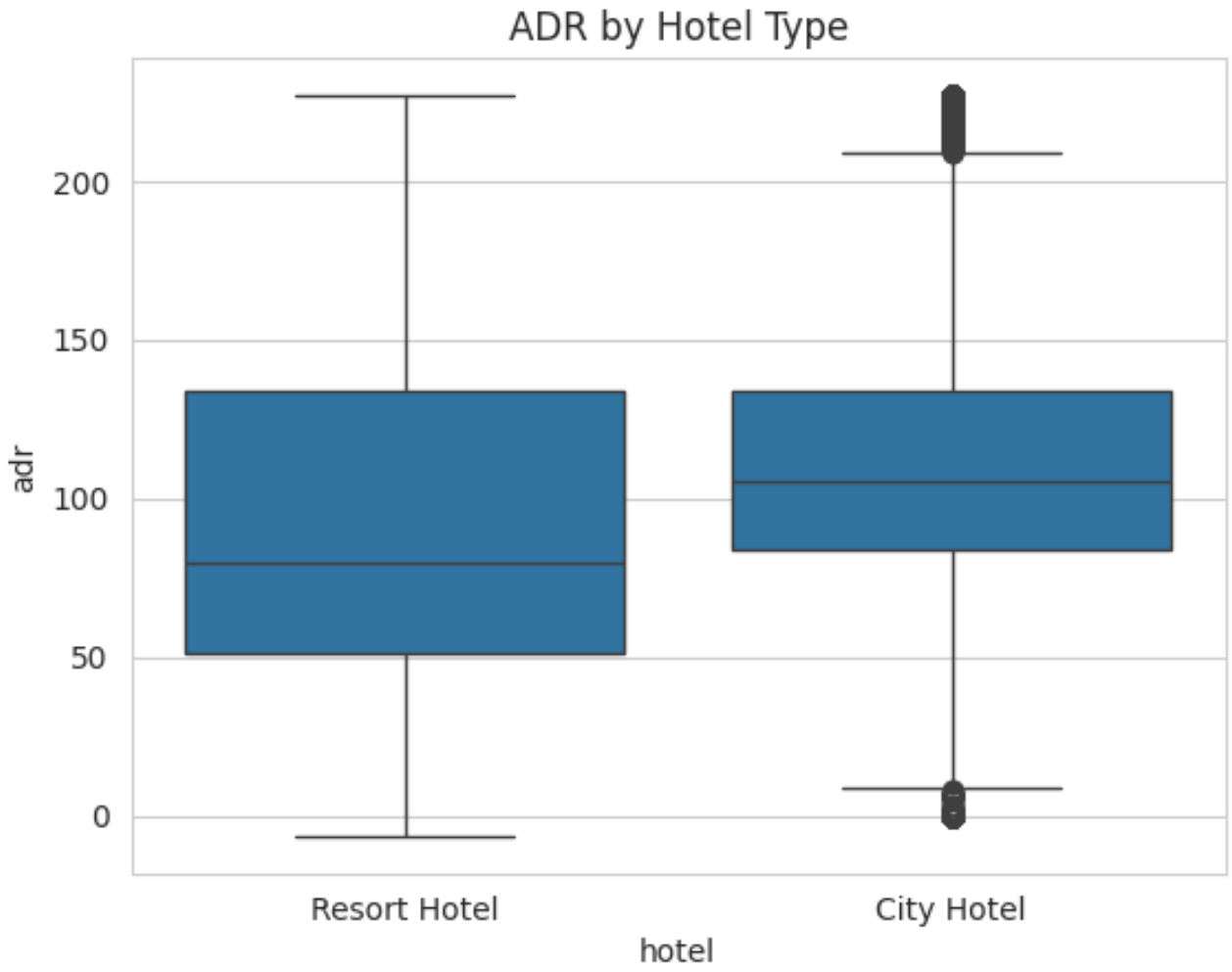
Market Segment Distribution



Finding

Certain market segments contribute significantly more bookings than others, indicating important customer acquisition channels.

ADR by Hotel Type



Finding

ADR varies between hotel types, suggesting different pricing strategies and customer preferences.

5. Feature Engineering and Encoding

Feature Engineering

Two new features were created:

Total_Guests = Adults + Children + Babies

Total_Nights = Weekday Nights + Weekend Nights

These features provide a better representation of customer size and stay duration.

Ordinal and Nominal Variables

Ordinal Variables

- meal

Nominal Variables

- hotel
- country
- market_segment
- distribution_channel
- customer_type
- reserved_room_type

Encoding Techniques

Label Encoding was applied to ordinal variables, while One-Hot Encoding was applied to nominal variables. These transformations prepare the dataset for future predictive modeling tasks.

6. Key Findings

1. City Hotels received more bookings than Resort Hotels.
2. ADR contained several outliers and was treated using Winsorization (capping).
3. Missing values were successfully handled using mean and mode imputation.
4. Duplicate records were removed to improve data quality.
5. Market Segment significantly influenced booking volume.

6. Correlation analysis revealed meaningful relationships among several numerical variables.
7. Feature engineering created useful variables such as Total Guests and Total Nights.

7. Business Conclusions

1. City Hotels experience greater booking demand and should receive increased operational focus.
2. High-performing market segments should be prioritized in future marketing strategies.
3. Understanding cancellation patterns can help improve revenue management and forecasting.
4. Monitoring ADR and customer booking behavior can support better pricing decisions.
5. Feature-engineered variables such as Total Guests and Total Nights provide valuable insights into customer behavior.

8. Conclusion

Exploratory Data Analysis was successfully performed on the Hotel Booking Demand Dataset. Data cleaning, missing value treatment, duplicate removal, outlier analysis, correlation analysis, feature engineering, and encoding techniques were applied successfully. The analysis revealed important insights regarding booking behavior, hotel performance, customer characteristics, and pricing trends. The prepared dataset is now suitable for future predictive modeling and business decision-making.